

Adithya S Nair

AI Engineer

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PROFILE

Full Stack AI Engineer and Consultant with around two years of production experience building and maintaining the core of Doctreen, a medical AI platform - LLM pipeline design, RAG workflows, prompt engineering, model evaluation, and FastAPI services across multiple providers. Skilled at building internal tooling and data workflows that cut costs and lift performance, from DICOM vision pipelines to a token-efficient serialisation format. Alongside this, led the ACM Student Chapter at Amritapuri as Chairperson, growing a 200+ member technical community, running national-level hackathons, and mentoring juniors in ML. Now pursuing MTech in CSE (AI/ML) while continuing to ship production AI systems and grow into technical leadership.

EXPERIENCE

Doctreen *AI Engineer* 06/2025 – 05/2026 | Montpellier, France (Remote)

- Develop and maintain AI features for a medical report-generation platform, integrating LLMs (OpenAI, Nebius, OVH) with prompt engineering and retrieval-augmented generation (RAG) to improve diagnostic accuracy and output quality.
- Designed and optimised end-to-end vision analysis pipelines for DICOM medical imaging across 10+ modalities, benchmarking model performance against radiologist-derived ground truth.
- Served as acting Product Manager, conducting market research on widespread modality usage, gathering user feedback, and authoring comprehensive Product Requirements Documents (PRDs).
- Built internal tooling and data-processing workflows, including a token-efficient serialisation format.

Tech: Python, FastAPI, LLM APIs, RAG, Prompt Engineering, Docker, CI/CD, Model Evaluation, Git, Jira, Agile, Linting, Testing

Doctreen, *AI Research Intern* 01/2025 – 06/2025 | Montpellier, France (Remote)

- Contributed to the foundational development of an AI-assisted medical report system
- Implemented security guardrails and input validation to harden the system against prompt injection, adversarial inputs, and to ensure healthcare data compliance.
- Developed a multi-model benchmarking system to evaluate LLM performance across all pipeline stages with parallel execution.

EDUCATION

MTech, Computer Science with Artificial Intelligence and Machine Learning 07/2026 – Present | Amritapuri, Kollam, India
Amrita Vishwa Vidyapeetham

- Research Focus: Agentic AI, Memory, Multi-Agentic System, LLM Architecture.

B-Tech, Computer Science with Artificial Intelligence and Machine Learning 09/2021 – 08/2025 | Amritapuri, Kollam, India
Amrita Vishwa Vidyapeetham | *Cgpa- 8.48*

- Coursework:* Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, DBMS, Operating Systems.
- SPARE:* Single-view Parameter-efficient Adapter for Radiology Reporting (Springer publication (PCCDA 2026))

SKILLS

Programming Languages: Python

AI / ML: LLM Integration & Prompt Engineering, Claude Skills, RAG, Embeddings, Deep Learning, Multi-Agent Systems, Model Evaluation & Monitoring

Frameworks: PyTorch, TensorFlow, Scikit-learn, FastAPI, LangChain, CrewAI, Django, React, Next.js, Streamlit, Gradio

Infrastructure: Docker, Git, REST APIs, MongoDB, MariaDB, Postman, Hugging Face, Jira & Confluence

Professional Values: Leadership | Results and Detail-Oriented | Diversity | Agile | Communication Skills | Continuous Learner | Initiator | Adaptable | Integrity

PUBLICATIONS

SPARE: Single-view Parameter-efficient Adapter for Radiology Reporting, PCCDA-2026, SPRINGER

- Engineered SPARE, a vision-language framework that connects the RAD-DINO vision encoder to BioGPT through a custom Semantic Alignment Adapter.
- Optimised the pipeline specifically for single frontal chest X-rays to support automated reporting in resource-limited and mobile clinical environments.
- Implemented Low-Rank Adaptation (LoRA) to reduce trainable parameters by 98%, successfully keeping memory consumption within consumer GPU budgets.
- Achieved a BERTScore F1 of 79.0% on the MIMIC-CXR dataset utilising just 0.8% of the training corpus, retaining 93.6% of state-of-the-art performance

PROJECTS

CodrantLabs  , Founders as a Service (FAAS)06/2026

CodrantLabs is a founder-led product and AI engineering studio for service businesses, startups, and enterprise teams that need software shipped, not strategy left in slides. We build full-stack web platforms, AI agents, RAG systems, and search visibility systems with fixed scope, direct senior engineering, and global delivery from Kerala.

Vidyapath  , LMS & School Management Platform for CBSE Schools04/2026 – Present

- Built a full-stack multi-school LMS targeting CBSE Class 10 & 12, with dedicated portals for teachers, admins, students, and parents covering assignments, grading, attendance, gradebooks, and timetable management in a single platform.
- Integrated a multi-model AI routing layer (Groq, Gemini) to power AI-assisted assignment generation and question banking, reducing manual teacher workload while keeping output pedagogically structured.
- Architected role-based auth with HMAC-signed JWTs and Supabase Postgres, supporting multi-school provisioning via a developer admin portal with usage analytics, audit trails, and school onboarding workflows.

Skills: Next.js 14, Supabase, Tailwind CSS, Prompt Engineering, RAG, Vectorising, Multi-model AI (Groq /Gemini), Zustand, Framer Motion, Open-Source

LensAI  , AI-Powered Screen Region Explainer02/2026 – 03/2026

- Built an MVP browser extension that lets users highlight any source code, diagrams, articles, UI, and get instant, context-aware explanations powered by LLM integration, solving the constant context-switching friction developers and researchers face daily.
- Designed the capture-to-explanation pipeline: selected regions are processed through an AI backend and returned as clear, structured breakdowns without ever leaving the current tab.

Skills: JavaScript, Browser Extension APIs, LLM Integration, Claude Code and Claude Skill development.

Namude Yatra 02/2025 – 03/2025

- Built a multi-agent travel planner that takes a destination and generates full day-by-day itineraries, handling research, scheduling, and recommendations through coordinated LLM agents using LangChain.
- Added a Streamlit frontend with interactive map visualisations (Pydeck) and a chatbot for follow-up questions and real-time trip adjustments.

Skills: LLM-based Agents, LangChain, Streamlit, Pydeck, Geopy, Generative AI.

OptiHire 10/2024 – 11/2024

- Built a web app that generates tailored cover letters and emails based on job descriptions, analyses resumes for keyword gaps, and tracks applications in one place, helping job seekers personalise their outreach without doing it manually every time.
- Integrated multiple APIs for job data scraping, LLM-based text generation, and resume parsing, with a Streamlit frontend for real-time feedback and visual analytics.

Skills: Streamlit, Web Scraping, LLM, NLP, Multi-API integration, Prompt Engineering, Business Intelligence.

SWC File Generation from MRI, Role - MRI Refining & Graph Generation08/2023 – 02/2024

- Developed AI software for generating SWC files to analyse human brain vascular flow from MRIs, focusing on refining MRI data and creating detailed graph representations for enhanced medical insights.

LEADERSHIP & COMMUNITY

ACM Student Chapter, Amritapuri2022 – Present | Kollam, Kerala

Member → AI Club Mentor → Chairperson → Advisory Council

Led the chapter as Chairperson for 1.5 years; organised national-level hackathons, coding competitions, workshops, and the annual tech fest. Mentored students in the AI Club on ML fundamentals and project development. Now serve on the Advisory Council, providing strategic guidance.

Vidyut Multi-Fest, Core Committee '24, Executive Member '232023 – 2024

Managed participant accommodations and logistics for one of Kerala's largest student-run multi-fests

ICPC Asia West Regional Finals, Overall Coordinator2022 – 2023

Coordinated event logistics, participant management, and on-ground operations for the ICPC regional finals hosted at campus.

Decoding AI, Student Social Responsibility (SSR) Project

Conducted an introductory AI awareness program at Sreyas Public School, Kottayam addressing the AI knowledge gap and fostering equitable access to technology education among school students.

CERTIFICATES

- Computer Vision 
 - DL for Images, Text with PyTorch
- Cybersecurity Essentials Badge 
- Data Visualisation: Empowering Business with Effective Insights 
- bit.ly/certificate-repo 

LANGUAGES

English: (Professional proficiency)

Malayalam: (Native proficiency)

Hindi: (Working proficiency)